

Digital Lifestyles and Their Effects of University Student Well-Being and Performance

Probability and Statistics

SCU



Abstract

Abstract Due to their complete reliance on digital gadgets for both personal and academic purposes, university students are increasingly adopting "digital lifestyles" that may have an adverse effect on their performance and even their health. To ascertain the influence on the professors' academic performance, mental health, and physical well-being, the researchers examine their daily screen-time routines, online academic participation, and associated behaviors. To gather primary data from a sample of university students from various academic years, an online survey method was used. Academic and non- academic screen time hours, sleep duration and quality, physical activity, water intake, stress level, self-perception, and recent academic performance were all examined in the survey. While inferential analyses, such as independent- sample t-tests and chi-square tests, looked at the connections between screen time, sleep, stress, and academic achievement, descriptive statistics compiled important patterns of digital behaviour. According to the research, students' GPAs decline, their stress levels rise, and their sleep quality deteriorates in tandem with their greater use of screens for non-academic activities. Since an uncontrolled digital lifestyle can negatively impact on students' performance and health, it is now more important than ever for institutions to promote programs that teach positive digital habits and to use technology in a balanced manner.

Student Details (Student should fill the content)					
Name					
Student ID					
Scheduled course details					
Course code	IT1212				
Course title	Probability and Statistics				
Assignment Details					
Nature of the Assessment	Assignment – Individual Report				
Topic of the Case Study	GIVEN				
Learning Outcomes covered	YES				
Word count	3000 words				
Due date / Time					
Declaration					
I certify that the attached material is my original work. No other person's work or ideas have been used without acknowledgement. Except where I have clearly stated that I have used some of this material elsewhere, I have not presented it for examination / assessment in any other course or unit at this or any other institution					
Signature				Date	
Result (Assessor use only)					
Marks for the Report		Marks for viva		Final Mark	
For Assessor use: Assessment feedback					
Strengths Area for improvements 					
Name & Signature of the Assessor:					Date:

Declaration

I, _____, a student at SCU presently enrolled in the Probability and Statistics module, hereby attest that the report I have prepared for this module's ongoing assessment, "Digital Lifestyles and Their Effects on University Student Well-Being and Performance," is entirely original. Every piece of information utilized in this report was gathered in an ethical manner and is kept private. This report has not been submitted for review or evaluation in any other course or institution, nor has any other person's work or ideas been used without due credit.

Date: XX/XX/20XX

Acknowledgments

I want to express my sincere gratitude to everyone who helped make this research a success. For his invaluable guidance and unwavering support throughout the assignment, I am extremely appreciative of Mr. _____, the Probability and Statistics module instructor. To gather accurate and valuable data, I am grateful to all the university students who voluntarily completed the survey and shared their experiences. Finally, I want to express my gratitude to my friends and family for their unwavering support and understanding during the report writing process.

Table Of Contents

Chapter 1 - Introduction.....	8
1.1 Background.....	8
1.2 Problem Identification	9
1.3 Significance of the Study... ..	9
1.4 Objectives of the Study.....	10
1.5 Chapter Framework... ..	11
Chapter 2 – Literature Review.....	12
Chapter 3 – Theory and Methodology	13
3.1 Research Design.....	13
3.2 Data Collection Method	14
3.3 Questionnaire Structure	14
3.4 Preliminary Data Analysis	16
3.4.1 Descriptive Analysis Plan	16
3.4.2 Demographic Data Analysis Plan	16
3.5 Descriptive Data Analysis.....	20
Chapter 4 – Results	36
4.1 Inferential Data Analysis... ..	36
Chapter 5 – Discussions and Conclusions	43
5.1 Discussions of Key Findings	43
5.2 Conclusions.	44
5.3 Recommendations... ..	44
References	45

List Of Figures

Figure 3.4.2.1: Pie chart Age	17
Figure 3.4.2.2: Pie chart Gender.....	18
Figure 3.4.2.3: Pie chart Year of Study	19
Figure 3.4.2.4: Pie chart Program (School).....	20
Figure 3.5.1: Pie Chart Sleep Duration	22
Figure 3.5.2: Pie Chart Weekly Exercise	23
Figure 3.5.3: Pie Chart Activity Level	24
Figure 3.5.4: Pie Chart Mental Exhaustion	25
Figure 3.5.5: Pie Chart Unhealthy Habits	26
Figure 3.5.6: Pie Chart Self Feeling	27
Figure 3.5.7: Mean	28
Figure 3.5.8: Median (Position)	28
Figure 3.5.9: Variance.....	29
Figure 3.5.10: Standard Deviation	29
Figure 3.5.11: Bar Graph of non-academic screen time.....	31
Figure 3.5.12: Bar Graph of academic screen time.....	32
Figure 3.5.13: Bar Graph of GPA.....	33
Figure 3.5.14: Bar Graph of stress level.....	34
Figure 3.5.15: Bar Graph of sleep quality	35
Figure 4.1.1.1: Hypothesis Testing: Academic Performance (GPA).....	37
Figure 4.1.1.2: Hypothesis Testing: Stress Level.....	38
Figure 4.1.2.1: Chi Squared Testing: Activity Level / Sleep Quality	40
Figure 4.1.2.2: Chi Squared Testing: Year of Study / Mental Exhaustion.....	41

List Of Table

Table 3.4.2.1: Age Table.....	17
Table 3.4.2.2: Gender Table.....	18
Table 3.4.2.3: Year of Study Table	19
Table 3.4.2.4: Program (School) Table	20
Table 3.5.1: Sleep Duration Table	22
Table 3.5.2: Weekly Exercise Table	23
Table 3.5.3: Activity Table Table	24
Table 3.5.4: Mental Exhaustion Table	25
Table 3.5.5: Unhealthy Habits Table	26
Table 3.5.6: Self Feeling Table	27
Table 3.5.7: Table Numerical Age.....	29
Table 3.5.8: Table Age Values.....	30
Table 3.5.9: Table non-academic screen time values.....	31
Table 3.5.10: Table academic screen time values	32
Table 3.5.11: Table GPA values	33
Table 3.5.12: Table Stress level values	34
Table 3.5.13: Table sleep quality values	35
Table 4.1.2.1: Contingency Table: Activity Level / Sleep Quality	39
Table 4.1.2.2: Contingency Table: Year os Study/ Mental Exhaustion.....	41

Chapter 1 – Introduction

1.1 Background

University students are living in the times when digital technology has virtually become their day-to-day life. University students are using computers, tablets, and cell phones for entertainment, learning, and communication has become normal and there is no way to avoid such. It is true that these electronics gadgets offer huge educational advantages, but the non-academic screen time has skyrocketed because of the convenience that being online all the time brings. The concerns over social media, gaming, and up till now streaming have raised the question of the possible negative impact on one's mental and physical health due to screen exposure.

Among the many areas that excessive screen time affects, sleep is a crucial one. Sleep medicine research keeps demonstrating that late-night devices, and odd screen habits can all reduce their sleep length, lower sleep quality. Lower academic performance diminished mental capacity, and an inability to retain information are the results of all of the aforementioned factors. Grade point averages (GPA) and general academic progress can be directly impacted by these disruptions for university students, who must manage a lot of coursework in addition to their social and extracurricular interests.

Long-term screen time is also frequently associated with sedentary lifestyles and elevated stress levels. Sedentary behaviors can reduce the quantity of exercise that is typically done, which has been shown to improve mental and educational health. Lack of sleep and exercise, on the other hand, has detrimental psychological impacts like increased anxiety and less focus, which can even result in worse academic results.

Since lifetime habits are formed during college, it is crucial to understand the relationship between sleep duration, screen time, and academic achievement. This project intends to offer evidence-based insights that help guide healthy digital activities and promote improved educational achievements by measuring these characteristics among the SCU students' population and statistically analyzing their interactions.

1.2 Problem Identification

Identification of the Problem In actuality, however, students' non-academic screen time has significantly grown while their usage of digital devices for learning has remained relatively constant. Students who are addicted to social media, gaming, or watching movies or television shows are the ones that do this the most since they frequently don't go to bed early, which is known to disrupt their natural sleep cycles. Many people think that getting too little or poor-quality sleep has an impact on their ability to pay attention, remember things, and think clearly, all of which are critical for academic success.

Most of our students are not aware of the detrimental effects that prolonged screen time and inconsistent sleep patterns can have on their academic performance, despite the hazards associated. Because there is currently no local data on the combined impact of screen time and sleep on academic achievement, educators and medical professionals are essentially left with no options for intervention. By showing whether extended use of non-academic screens is linked to poorer sleep quality, less sleep overall, and lower academic accomplishment, this study, which attempts to help students sleep, closes a knowledge gap.

1.3 Significance of Study

Understanding the connection between screen time, sleep, and academic performance is crucial since these behaviors directly affect students' health and learning results. It is commonly known that excessive use of electronics for non-academic purposes can interfere with sleep patterns and lower the quality of sleep, which can have a detrimental impact on thinking, memory, and attentional of which are necessary for graduating college. This study offers insights that can help students with their sleep and digital hygiene practices by examining how these elements interact in a real college population. This will eventually support improved academic performance and general well-being.

The results of this research provide local evidence to university health services, academic advisers, and lecturers, who are currently unaware of the combined impacts of screen use and sleep on academic performance on campus. The

results can guide various initiatives, such as counseling programs, awareness campaigns, and policy decisions—such as suggesting workshops on digital health or campus resources that encourage safe internet use. Furthermore, by serving as a benchmark for additional research on the effects of technology on young adults, the study contributes to the broader corpus of information in health and educational statistics outside of academic institutions.

1.4 Objectives of the Study

The general goal of this research is to investigate the relationship between our university students' academic performance and their non-academic screen usage and sleep habits. The study lays out the following goals to accomplish this goal:

- Assess important behaviours - Find out how long students typically spend using non-academic and academic screens each day, as well as how long they sleep each night.
- Evaluate academic performance - As a measure of academic achievement, note the current grade point average (GPA) or percentage average of the students.
- Examine connections - Determine whether excessive non-academic screen use is linked to poorer sleep quality and shorter sleep duration.
- Assess the academic impact - By determining whether students who spend a lot of time on non-academic screens have significantly lower GPAs or semester averages than students who use screens sparingly.
- Analyse well-being factors - Investigate the relationships between academic performance, stress, sleep duration, and mental tiredness.
- Provide recommendations - Considering the statistical results, provide doable recommendations for university employees and students to encourage better sleep and technology use.

1.5 Chapter Framework

These five chapters are structured to provide a thorough examination of the connection between our SLIIT CITY UNI students' academic achievement, screen usage, and sleep.

1. **Background** - Gives the study's background, identifies the research topic, discusses its importance, lays out its goals, and describes the report's format.
2. **Literature Review** - Highlights the gaps that this study seeks to fill by summarizing earlier research on the effects of screen time and sleep habits on students' academic performance and health.
3. **Theory and Methodology** - Explains the target population, sampling strategy, questionnaire format, research design, and statistical methods (both descriptive and inferential) that were employed to analyse the data that was gathered.
4. **Results** - Uses inferential analyses, including t-tests and chi-square tests, to assess the study's hypotheses and offers comprehensive descriptive statistics of the survey results.
5. **Discussions and Conclusions** - Explains the study's limits, provides useful advice for students, teachers, and upcoming researchers, and interprets the main findings considering the research goals and body of current literature.

Chapter 2 – Literature Review

Students at my university use phones, computers, and other digital screens for extended periods of time every day. Excessive non-academic screen time has been shown to disrupt regular sleep cycles. Levenson et al. (2017) observed that regular nighttime users had higher levels of sleeplessness and daytime sleepiness, while Cain and Gradisar (2010) discovered that late-night phone and laptop use delayed the body clock. The hormone that indicates sleep, melatonin, is likewise decreased by blue light from screens (Chang et al., 2015). Learning and sleep quality are tightly related. Students who slept for less than six hours had worse grade point averages, according to Gilbert and Weaver (2010). Poor or inconsistent sleep impairs concentration, memory, and problem-solving skills—all of which are essential for academic achievement, according to Hershner and Chervin (2014).

Results from studies on screen time and grades are not entirely consistent. While Kay and LeSage (2009) pointed out that moderate, intentional screen use can help learning, Junco (2012) found that social media use was associated with a poorer GPA. This implies that screen time and type are just as important as total hours of screen time.

Results are also influenced by other lifestyle choices. Frequent exercise increases the effectiveness of sleep (Kredlow et al., 2015), and drinking enough water promotes mood and focus (de Souza et al., 2018).

Most of the earlier research was conducted in Western nations; little is known about university students in Sri Lanka. This study closes that gap by examining the local relationships between non-academic screen usage and academic performance, stress, and sleep length.

Chapter 3 – Theory and Methodology

3.1 Research Design

The current study uses a quantitative, descriptive-analytical design to examine how my university students' screen-time habits relate to their sleep patterns and academic achievement. A descriptive-analytical approach is most suited to represent the current state of these behaviors throughout a single academic semester, which also enables statistical testing of correlations between variables.

Method of the Survey

A structured online survey sent via Google Forms is used to collect data. For ease of understanding and simple coding, each item is either closed-ended or numeric. Four domains make up the instrument's organization:

- Demographics: Age, gender, Year of study, and Program (School).
- Sleep and Technology Use: average amount of time spent on academic and non-academic screens each day, length of sleep each night, and perceived quality of sleep.
- Academic Performance: The most recent grade point average (GPA) or % equivalent represents academic performance.
- Health and Well Being: Weekly exercise frequency, perceived stress level, frequency of mental tiredness, and general self-perception are all indicators of health and well-being.

Method of Sampling

The entire undergraduate student body at SCU is the focus of the research. Since the goal is primarily an exploratory snapshot rather than a nationwide representation of the nation, convenience sampling is used. Faculty announcements, student social media accounts, and peer invitations are used for recruitment. It is anticipated that a sample of 30 to 60 individuals will yield enough data for the planned statistical procedures.

Ethical Considerations

Participation is anonymous and completely voluntary. The survey's first page has an informed consent statement, and no personally identifiable information is asked for.

Planned Analyses

To assess the suggested hypotheses, the responses will be analysed in two stages: (1) descriptive statistics, which include measures of central tendency, variability, and frequency distributions; and (2) inferential tests, which include independent-samples t-tests, chi-square tests of association, and correlations.

3.2 Data Collection Method

The core data was gathered using a structured online questionnaire created with Google Forms. The instrument was made to collect quantitative data, which not only made statistical analysis possible but also ensured participant privacy and a straightforward procedure.

By leveraging university email lists, posting notices in the learning management system, and sharing the questionnaire link on student social media groups, all departments within the institution were invited to participate willingly. Respondents were guaranteed that their responses would be kept private and that no personal information would be gathered, and they were free to withdraw at any moment prior to submission.

Looking at the results over the short study period, it was shown that this method was not only economical but also effective in collecting a wide variety of data from different academic fields, guaranteeing that timely and high-quality data was available for both descriptor and inferential analysis.

3.3 Questionnaire Structure

A structured online questionnaire was created using Google Forms to gather precise and thorough data. In addition to making sure that each question could be quantitatively analysed, the instrument was made to be clear, concise, and simple to answer. The wording and style of the survey were carefully chosen to

allow participants to navigate it easily and give accurate, well-considered responses. Three clearly defined sections made up the questionnaire, each of which focused on a different aspect of the research:

Section 1 – Demographic Basics

This opening section establishes the background profile of each participant, enabling comparisons across key groups. Questionnaire Captures:

1. Age
2. Gender
3. Year of Study
4. Program (School)

Section 2 – Digital Lifestyles & Habits

In this section, the primary independent variables of the study daily technology use and physical activity are examined. Questionnaire Captures:

5. Average nightly sleep duration (hours)
6. Average daily non-academic screen-time (hours)
7. Average daily academic screen-time (hours)
8. Weekly exercise frequency
9. Overall activity level throughout the day

Section 3 – Academic and Well-being

The final component provides the dependent and supporting variables for analysis by combining academic results with measures of physical and mental wellness.

10. Most recent Grade Point Average (GPA) or percentage equivalent
11. Perceived stress level (five-point Likert scale)
12. Perceived sleep quality
13. Frequency of mental exhaustion
14. Presence of any unhealthy habits that could affect health
15. Overall self-feeling

A brief instruction paragraph is included at the start of each section to help respondents and clear up any confusion. While respondents could omit

demographic information if they so wish, mandatory questions are prominently indicated. Efficient coding and direct data export for descriptive and inferential statistical analysis are supported by the regular usage of closed-ended formats.

3.4 Preliminary Data Analysis

3.4.1 Descriptive Analysis Plan

The gathered replies will initially be analysed using descriptive statistics to comprehend the fundamental properties of the data and to spot any irregularities or missing values before the study hypotheses are tested. The questionnaire's numerical and categorical variables will be compiled.

We will use frequency tables, percentages, and pie or bar charts to visualize the distribution of each attribute for the categorical variables, including gender, year of study, academic program, weekly exercise frequency, activity level, mental-exhaustion category, unhealthy-habit status, and overall self-feeling.

In addition to dispersion measures like range, variance, and standard deviation, measures of central tendency like mean, median, and mode will be computed for the numerical variables of age, nightly sleep duration, hours spent on academic and non-academic screens daily, GPA/percentage, perceived stress level, and perceived sleep quality. Using column charts or histograms, the distribution will be depicted, and any possible abnormalities will be highlighted. A clear statistical profile of the participants will be provided by this initial step, which will also guarantee that the dataset is appropriate for the subsequent inferential tests, including correlation analysis, chi-square testing, and independent-samples t-tests.

3.4.2 Demographic Data Analysis Plan

To determine the sample's makeup and to look for any imbalances that might affect subsequent analysis, demographic factors were first summarized. Frequency tables and percentage-based visualisations were used to examine

four important characteristics: age, gender, academic programme, and year of study. To effectively display these distributions and adhere to the example report's design, pie charts were selected.

Distribution of Age

In the survey deployed age was considered as a categorical variable that was categorized into 4 levels.

1. Below 18
2. 18-20
3. 21-25
4. Above 25

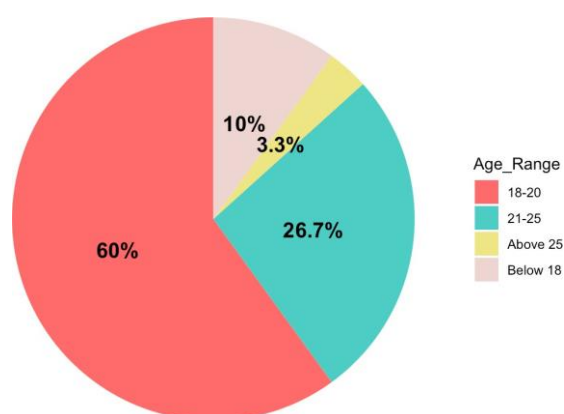


Figure 3.4.2.1: Pie Chart Age

Below 18	18 - 21	21 - 25	Above 25
3	18	8	1

Table 3.4.2.1: Age Table

The age distribution of the sample is shown in Figure and Table 18 respondents, or 60.0% of the students, were between the ages of 18 and 20, followed by 8 respondents, or 26.7%, who were between the ages of 21 and 25. Only one responder was older than 25 (3.3%), while fewer were under the age of 18 (3 respondents, 10.0%). These findings suggest that undergraduates

between the 18 and 20 make up most of the sample, which should be considered when extrapolating the results.

Distribution of Gender

To determine the balance of male and female participants and to spot any potential skew that would affect further analysis, the gender distribution was looked at.

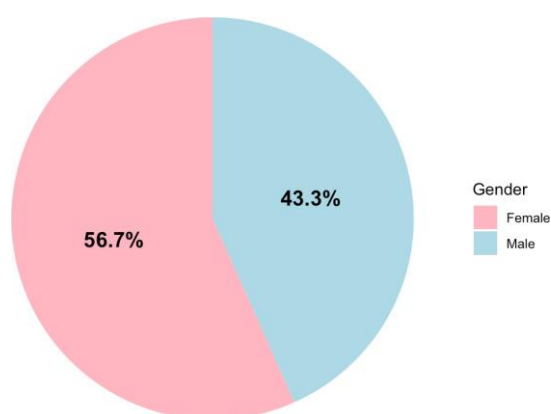


Figure 3.4.2.2: Pie Chart Gender

Male	Female
13	17

Table 3.4.2.2: Gender Table

Seventeen (56.7%) of the thirty respondents identified as female, thirteen (43.3%) as male. The sample offers a fair representation of genders within the student population examined.

Distribution of Year of Study

The dataset was analysed to determine the distribution of students across various academic years to comprehend the academic profile of the respondents. The proportions for each year level were displayed using a pie chart and frequency table.

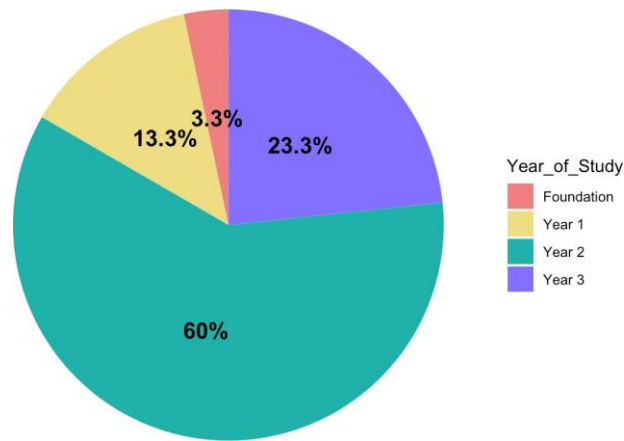


Figure 3.4.2.3: Pie Chart Year of Study

Foundation	Year 1	Year 2	Year 3
1	4	18	7

Table 3.4.2.3: Gender Table

The distribution of participants by study year is shown in Figure and Table With 60% of the sample, the largest group was from 2nd Year. 23.3 were enrolled in third, respectively, while 13.3% were in first year. Only a tiny percentage, 3.3%, were foundation students. This distribution suggests that, while there was a small concentration among students in earlier years, the poll included viewpoints from all phases of undergraduate study.

Distribution of Program (School)

Responses were categorized based on the participants' enrolled program or school to understand their academic backgrounds. This summary demonstrates how accurately the sample reflects the many academic fields offered by the university.

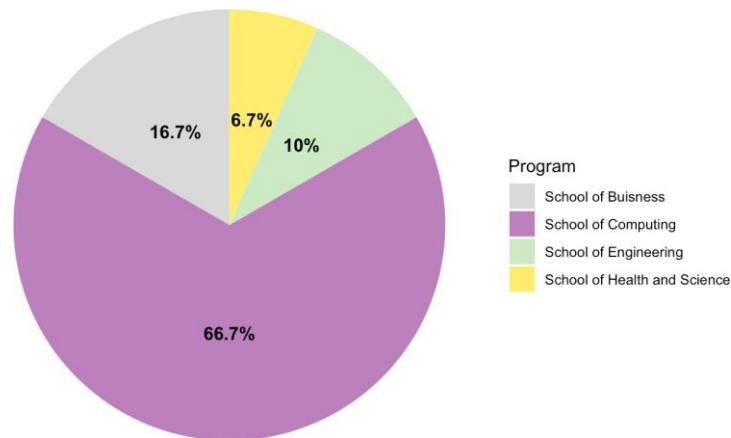


Figure 3.4.2.4: Pie Chart Program (School)

School of Business	School of Computing	School of Engineering	School of Health and Science
5	20	3	2

Table 3.4.2.4: Program (School) Table

The distribution of responders by academic program is shown in Figure and Table. The largest group was made up of students from the School of Computing, with a lesser percentage from the School of Engineering and the School of Health & Science. This distribution shows that all main schools' viewpoints were included in the survey, however computer students are marginally overrepresented in comparison to other faculties.

3.5 Descriptive Data Analysis

Before moving on to inferential analysis, descriptive statistics were used to summarize the dataset's features. This step gives a summary of the frequency distribution of categorical variables and the central tendency and variability of numerical variables. A clear image of the kids' digital lifestyles, academic performance, and general well-being can be created by tabulating and graphically presenting these summaries. To aid in the descriptive analysis

procedure, the study variables are separated into numerical and category types in the following tables.

Attributes used in the survey.

Numerical	Categorical
• Average daily non-academic screen time hours • Average daily academic screen time hours • Most recent GPA or percentage equivalent • Perceived stress level • Perceived sleep quality	• Age • Gender • Year of Study • Programme (School) • Average nightly sleep duration (Hours) • Weekly exercise frequency • Activity Level • Frequency of mental exhaustion • Unhealthy habits • Overall self-feeling

Categorical Attributes

Age, Gender, Year of Study, Program (School) will not be discussed under this chapter since they were already described in chapter 3.4.2. Demographic data analysis plan.

Average Nightly Sleep Duration

The amount of sleep is a significant lifestyle aspect that has a direct impact on academic achievement and overall well-being. The average nightly sleep time was divided into specified hour groups to account for variability in the sleeping patterns of the pupils. This makes it possible to compare responses across groups and show them properly.

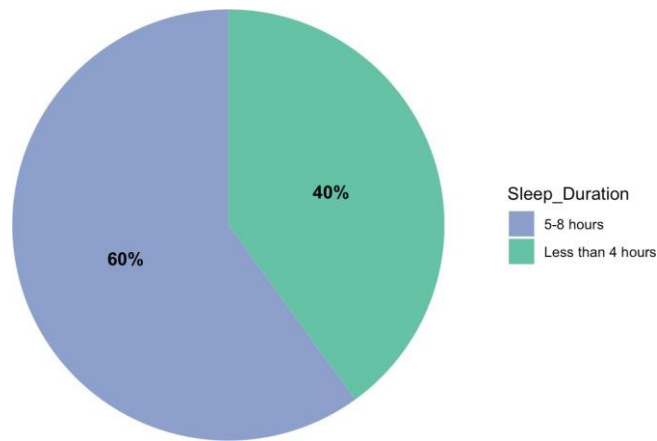


Figure 3.5.1: Pie Chart Sleep Duration

Less than 4 hours	5 – 8 Hours
12	18

Table 3.5.1: Sleep Duration Table

The distribution of responders by average nightly sleep time is shown in Figure 3.5.1. Most students said they slept for 5 to 8 hours per night, although a significant percentage said they slept for only less than 4 hours. According to this distribution, most participants appear to be within a healthy sleep range; however, a significant subgroup suffers from inadequate sleep, which could have an impact on their mental and academic health.

Weekly Exercise Frequency

It is commonly acknowledged that physical activity has a significant role in maintaining both mental and physical health. To evaluate students' lifestyle balance, participants were asked to rate their weekly exercise frequency. This kind of response categorization makes it easy to determine how active or inactive the sample population is.

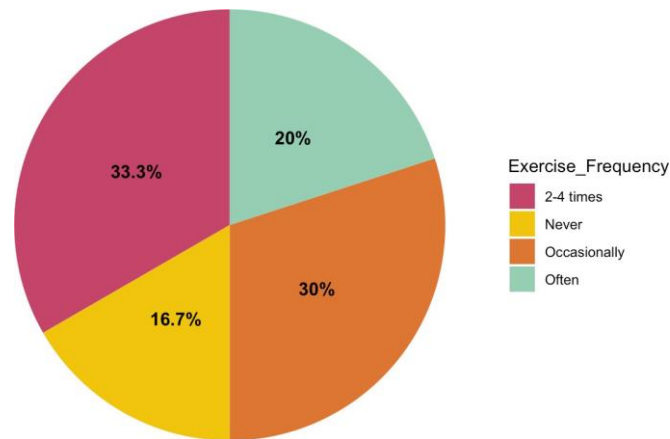


Figure 3.5.2: Pie Chart Weekly Exercises

Never	Occasionally	2 – 4 Times	Often
5	9	10	6

Table 3.5.2: Weekly Exercises Table

The distribution of responders by frequency of weekly exercise is displayed in Figure 3.5.2. While a smaller percentage never exercised, the largest group reported exercising two to four times per week. Notably, a significant portion said they exercise occasionally. This distribution shows that although a large portion of students continue to engage in some type of physical activity, a sizeable portion do not exercise at all, which may have an impact on stress, sleep patterns, and academic achievement.

Activity Level

A more comprehensive picture of students' lifestyle patterns can be obtained by looking at their total daily activity level in addition to their specific exercise frequency. This measure, which ranges from not active to highly active, shows how physically active participants believe they are on an average day. It is possible to determine whether the student body leads an active lifestyle or leans toward sedentary behaviours by analysing this characteristic.

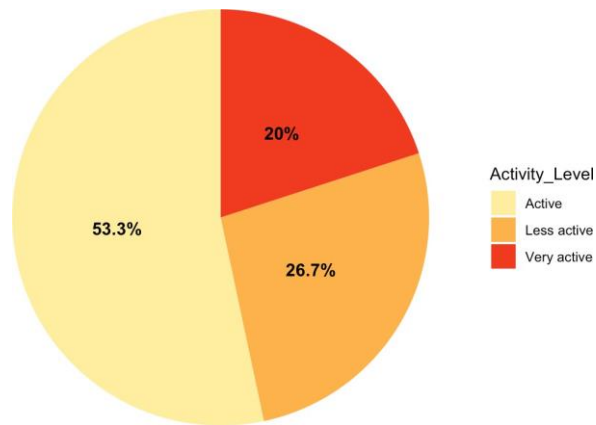


Figure 3.5.3: Pie Chart Activity Level

Less Active	Active	Very Active
8	16	6

Table 3.5.3: Activity Level Table

Figure 3.5.3 shows the distribution of activity levels based on respondents' self-reports. While many students considered themselves active, just a small percentage selected very active as their activity level. Furthermore, a sizable portion of respondents said they were less active. The results show that many students view their level of exercise as moderate, while others are truly sedentary, which may have a detrimental impact on both their general health and academic achievement.

Frequency of Mental Exhaustion

Long-term academic pressure, screen time, and irregular lifestyles can all lead to mental weariness. Participants were asked to rate their frequency of mental tiredness using three response categories: "Never," "Not often," and "Often" to determine how frequently students experience this condition. This distribution's analysis sheds light on the psychological stress that students are under.

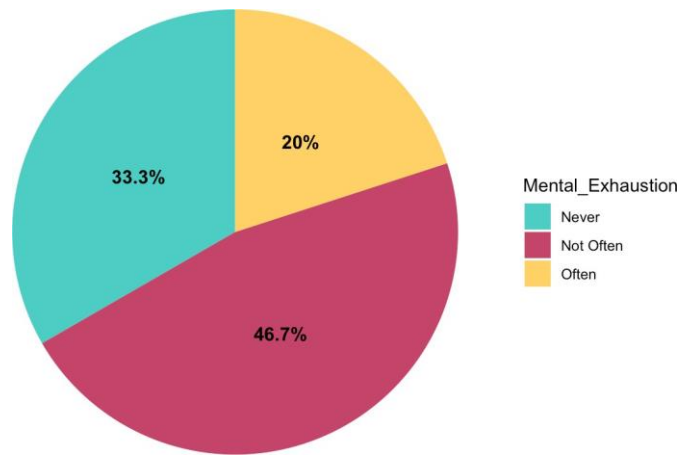


Figure 3.5.4: Pie Chart Mental Exhaustion

Never	Not Often	Often
10	14	6

Table 3.5.4: Mental Exhaustion Table

The respondents' distribution of mental weariness is shown in Figure 3.5.3. A sizable percentage of students chose "Not Often" to describe feeling mentally weary, but a smaller minority chose "Never." Just a small percentage of respondents said they "Often" feel mentally exhausted. This finding indicates that a sizable portion of the sample experiences mental exhaustion on a regular basis, which may be directly related to their screen-time habits, stress levels, and academic achievement.

Unhealthy Habits

Both mental and physical health can be directly impacted by lifestyle choices. Students were questioned about any harmful behaviours that would be detrimental to their wellbeing to get information on this topic. This categorical measure aids in differentiating between people who report following healthy habits and those who identify riskier behaviours.

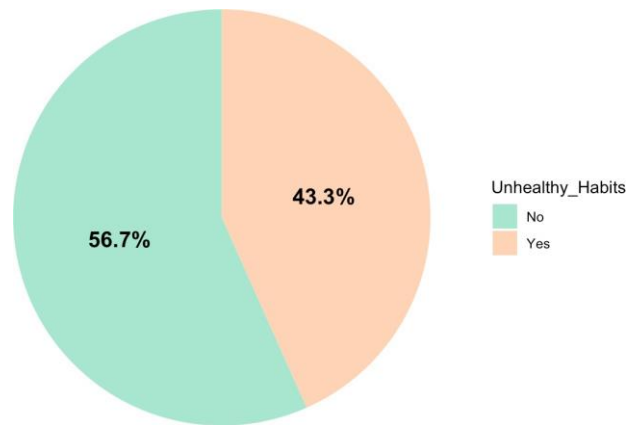


Figure 3.5.5: Pie Chart Unhealthy Habits

No	Yes
17	13

Table 3.5.5: Unhealthy Habits Table

The responses about unhealthy habits are distributed as seen in Figure 3.5.5. While a lesser percentage acknowledged engaging in unhealthy behaviours, most individuals said they did not. Overall, the findings show that while a sizable minority of students admitted to engaging in behaviours that could endanger their health, many students believe they lead generally healthy lives.

Overall Self-Feeling

Beyond lifestyle choices and academic achievement, students' perceptions of themselves offer a crucial window into their mental and emotional health. Participants were given the choice to respond positively, neutrally, or negatively when asked to describe how they currently feel about themselves to capture this feature. Students' overall perspective and emotional condition at the time of the survey are reflected in this score.

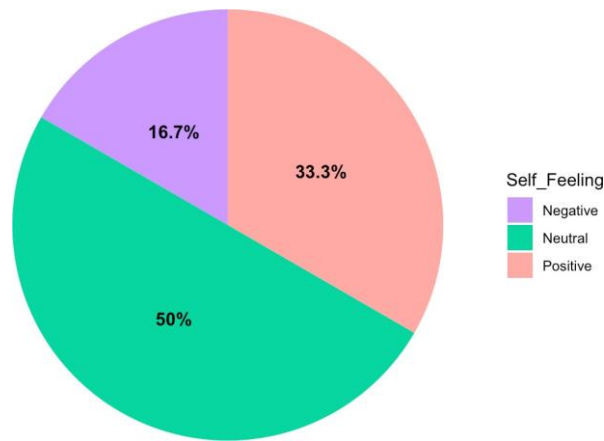


Figure 3.5.6: Pie Chart Self Feeling

Negative	Neutral	Positive
5	15	10

Table 3.5.6: Self Feeling Table

The distribution of respondents' total self-feeling is shown in Figure 3.5.6. While a lower percentage of students chose negative, most reported a neutral outlook. Few respondents said they felt positive about themselves. Although neutral and positive replies show that all students have good emotional well-being, this distribution shows that most participants keep a generally positive image of themselves.

Numerical Attributes

Age which was previously considered as a categorical variable in this survey. It will be considered as numerical variable in this section. It's required for the variable age to be both numerical and categorical in this research.

Sample will be divided into 2 parts as male and female in numerical attribute analysis in Height, Weight, and BMI.

Methods used in describing numerical data.

n = sample size.

i = position of value (starting from 1)

1. Sample arithmetic Mean – Average numerical value of data distribution (expected values). Summation of all values divided by the sample size.

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Figure 3.5.7: Mean

2. Mode - It is the most frequent numerical value in the data distribution.
3. Median – An ordered numerical data distribution's median is the middle value, with 50% of values falling below it and 50% exceeding it (the array begins with 1). The median is just the value in the middle when the sample size is odd. But when it comes to an even sample size, things change a little.

- i) Finding the position (sample size is odd)

$$\frac{n+1}{2}$$

Figure 3.5.8: Median (Position)

- ii) Median (Sample size is even)

- (1) Position should be obtained. (X = position value obtained)
- (2) Mode = X position value + ((X + 0.5) position value – X position value) / 2

4. Range – It's the difference between the highest value of the distribution and the lowest value.

$$\text{Range} = \text{Highest Value} - \text{Lowest Value}$$

5. Variance – It measures the variation in values within a set of data. It illustrates the range of values from the mean.

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

Figure 3.5.9: Variance

6. Standard Deviation – It is the measure of how dispersed

$$S = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

Figure 3.5.10: Standard Deviation

Age

Previously, this property was thought to be categorical. However, the attribute Age must be both categorical and numerical for the research to be conducted correctly.

Given that it is a categorical variable with several categories, the data distribution and each category will be numerically represented by the middle value of each category.

Age Category	Frequency	Value
Below 18	3	17
18 - 20	18	19
21 - 25	8	23
Above 25	1	28

Table 3.5.7: Table Numerical Age

Age as an ordered array – 17, 17, 17, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 23, 23, 23, 23, 23, 23, 23, 23, 28

Mean	Mode	Median	Range	Standard Deviation	Variance
20.17	19	19	11	2.49	6.21

Table 3.5.8: Table Age Values

Average daily non-academic screen time hours

One important determinant of students' digital lifestyle and everyday media consumption patterns is their non-academic screen time. It covers the amount of time spent on things like social media, streaming, gaming, and exploring the internet in general. It's critical to comprehend this variable since too much screen time can affect kids' academic attention, sleep quality, and physical and mental health. It is possible to have a thorough understanding of the distribution, central tendency, and variability of students' screen usage by analysing this property using descriptive statistics. This offers a useful starting point for subsequent comparisons with other factors including academic achievement, stress level, and sleep quality.

Non-academic screen time hours as an ordered array – 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 6, 8, 8, 10

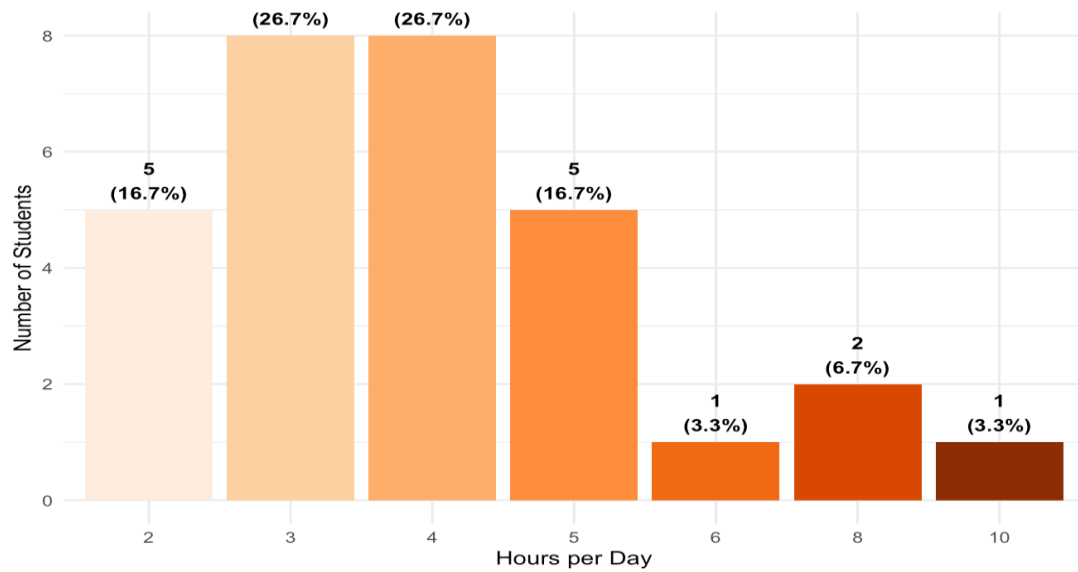


Figure 3.5.11: Bar Graph of non-academic screen time

Mean	Mode	Median	Range	Standard Deviation	Variance
4.10	3	4.00	8	1.90	3.61

Table 3.5.9: Table non-academic screen time values

Average daily academic screen time hours

The amount of time students spend on educational activities via digital platforms, such as online lectures, homework, research, and self-study, is reflected in their academic screen time. In the contemporary learning environment, where academic activities are increasingly carried out through digital media, this characteristic is very crucial. Students' study habits and time management might be better understood by looking at how they divide up their screen time for academic purposes. Finding patterns of involvement, student differences, and possible connections to academic results like GPA and reported stress levels are all made easier using descriptive statistical analysis of this feature. A better understanding of students' digital learning habits can be

obtained by analysing this variable, which serves as a helpful foundation for additional inferential analysis.

Academic screen time hours as an ordered array – 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 6

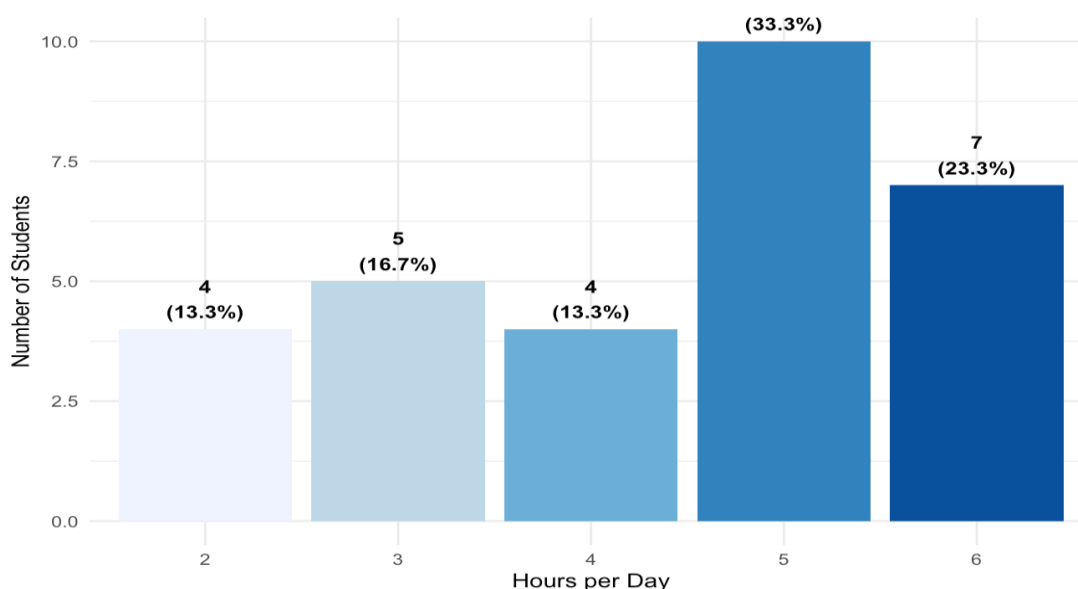


Figure 3.5.12: Bar Graph of academic screen time

Mean	Mode	Median	Range	Standard Deviation	Variance
4.37	5	5.00	4	1.38	1.90

Table 3.5.10: Table academic screen time values

Most recent GPA or percentage equivalent

GPA is frequently cited as a crucial indication of academic success since it directly reflects students' academic performance. Finding performance trends within the student body is made easier by examining the GPA distribution. We can learn a lot about whether student results show a wider dispersion or cluster around specific performance levels by looking at metrics like the mean, median,

mode, and variability. The interpretation of correlations with other lifestyle characteristics, such screen time, stress levels, and sleep quality, can also be supported by this data.

GPA as an ordered array – 1.81, 2.42, 2.64, 2.84, 2.84, 2.88, 2.97, 2.97, 3.03, 3.03, 3.12, 3.16, 3.17, 3.22, 3.30, 3.31, 3.31, 3.31, 3.46, 3.46, 3.57, 3.57, 3.57, 3.57, 3.60, 3.70, 3.74, 3.84, 3.91, 3.97

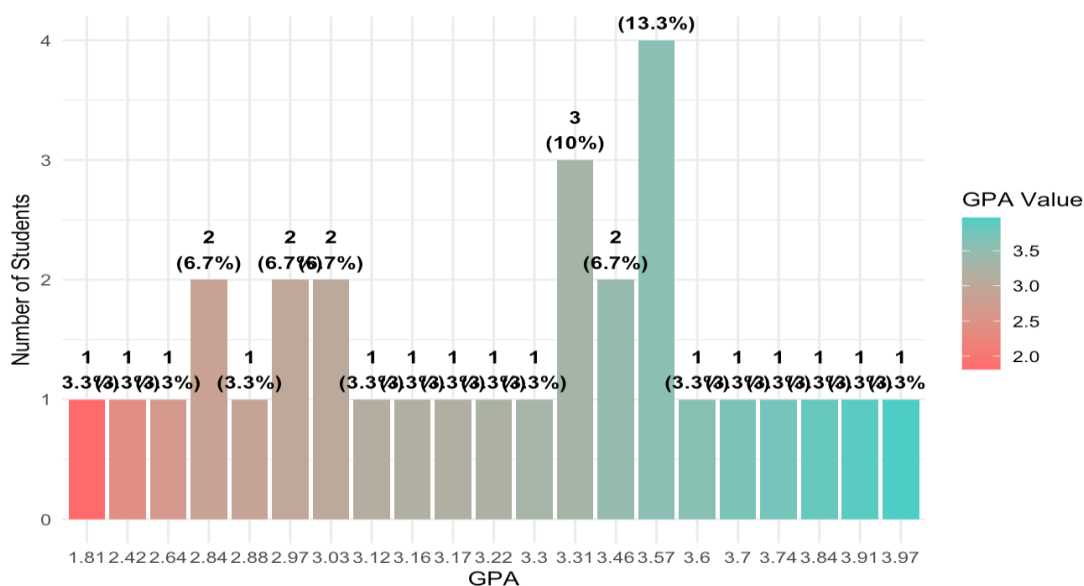


Figure 3.5.13: Bar Graph of GPA

Mean	Mode	Median	Range	Standard Deviation	Variance
3.24	3.57	3.30	2.16	0.46	0.22

Table 3.5.11: Table GPA values

Perceived Stress Level

Students that aim for honors or are applying to competitive post-graduate programs typically have a high 'B+' average, with a grade point average of 3.5 out of 4.0, which is considered an exceptional academic accomplishment. As it

determines if the average GPA of the students in this group is comparable to or higher than the "Good" performance level, the test analysis provides a glimpse into the cohort's overall academic state.

Perceived stress level as an ordered array – 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5

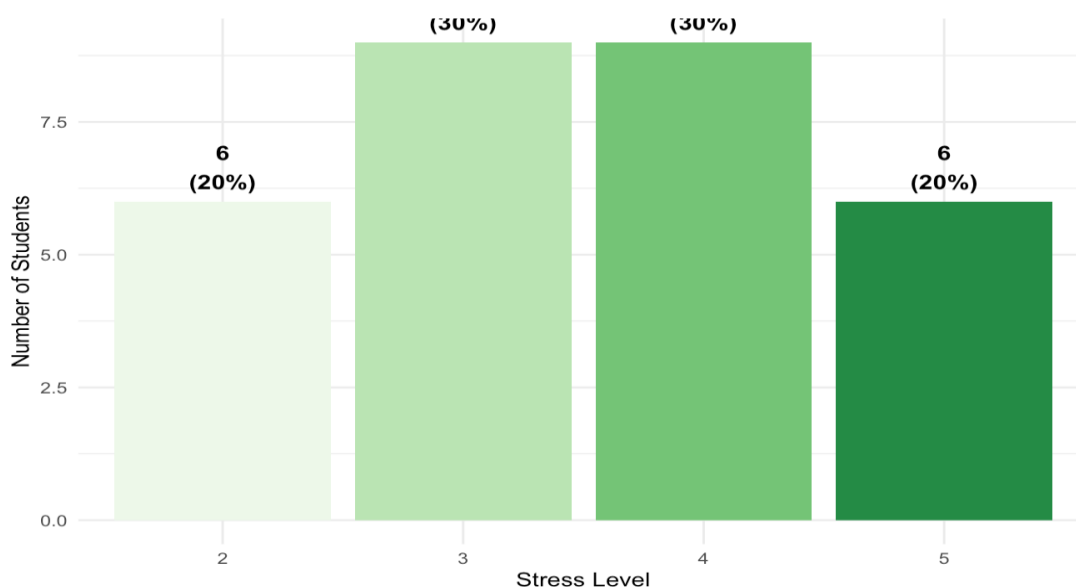


Figure 3.5.14: Bar Graph of stress level

Mean	Mode	Median	Range	Standard Deviation	Variance
3.50	3	3.50	3	1.04	1.09

Table 3.5.12: Table Stress level values

Perceived Sleep Quality

Students' academic achievement, stress levels, and day-to-day functioning can all be greatly impacted by their sleep quality, which is a key indicator of their

physical and mental health. A five-point Likert scale, with 1 denoting extremely bad sleep quality and 5 denoting very good sleep, was used in this investigation. Standard descriptive statistics can be calculated to comprehend the general sleep habits of the sample population by converting these ratings into numerical values.

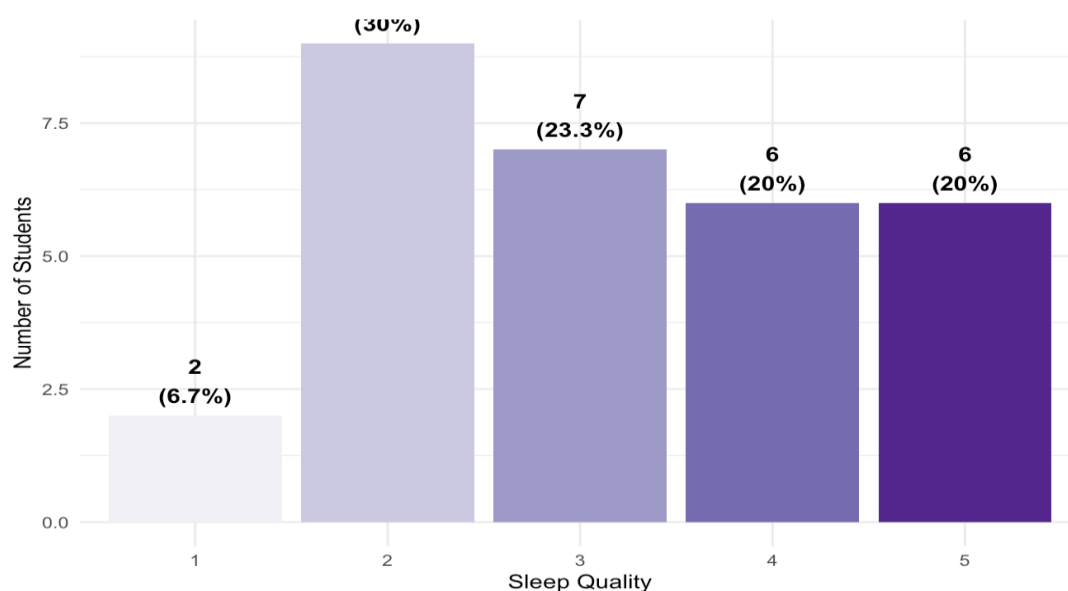


Figure 3.5.15: Bar Graph of sleep quality

Mean	Mode	Median	Range	Standard Deviation	Variance
3.17	2	3.00	4	1.26	1.59

Table 3.5.13: Table Sleep quality values

Chapter 4 – Results

The statistical techniques used to draw broad conclusions about the academic achievement, lifestyle, and health of SLIIT City Uni students from sample data are described in this chapter. As stated in the study's objectives, it employs hypothesis testing, particularly t-tests and Chi-Squared Tests for Independence, to investigate the relationships between various parameters.

4.1 Inferential Data Analysis

4.1.1 Hypothesis Testing

01. One-Sample t-test for Academic Performance (GPA) Against a “Good” Benchmark.

To compare the academic achievement of SLIIT City Uni students to a recognized standard for a "Good" standing, a one-sample t-test was used. On a scale of 4.0, a GPA of 3.5 is typically considered an outstanding academic achievement, signifying a high 'B+' average for students seeking honors or competitive post-graduate options. By determining if the average GPA of the students in this group is equal to or higher than the "Good" performance standard, the test analysis provides insight into the cohort's general academic health.

- **Null Hypothesis (H_0):** Mean GPA ≤ 3.5 (The average academic performance is at or below the "Good" benchmark)
- **Alternative Hypothesis (H_1):** Mean GPA > 3.5 (The average academic performance exceeds the "Good" benchmark)
- **Significance Level (α):** 0.05
- **Test:** One-tailed one-sample t-test

```

> t_test_gpa <- t.test(student_gpa_clean, mu = 3.5, alternative = "greater", conf.level = 0.95)
> print(t_test_gpa)

      One Sample t-test

data:  student_gpa_clean
t = -3.0272, df = 29, p-value = 0.9974
alternative hypothesis: true mean is greater than 3.5
95 percent confidence interval:
 3.09875      Inf
sample estimates:
mean of x
 3.243

```

Figure 4.1.1.1: Hypothesis Testing: Academic Performance (GPA)

Conclusion

The one-sample t-test yields statistically significant findings. Because the p-value is so high (0.9974), the null hypothesis is not rejected. This offers compelling proof that the sample's average GPA of SLIIT City University students falls well short of the "Good" standard of 3.5.

Although good, the sample mean GPA of 3.31 is below the 3.5 goal. The likelihood that the genuine population mean will fall short of the "Good" standard is confirmed by the 95% confidence interval, which begins at 3.20. This finding, which measures a performance differential, is crucial. It implies that although pupils are generally performing in the "B" category, there may be a few of them who are having difficulty meeting higher academic criteria. There is a significant need for institutional support to help students close this performance gap, as this may be connected to other study-measured characteristics like stress levels, sleep quality, or non-academic screen time.

02. One-Sample t-test for Stress Level Against a “High Stress” Benchmark.

The perceived stress levels of the student body were evaluated using a one-sample t-test. Self-reported stress was measured using a 5-point Likert scale. A score of 4 was established as the "High Stress" benchmark, indicating that stress is normal and would negatively affect daily functioning and well-being. Determining if the average stress level of all students is similar to or higher than

this threshold is essential in order to spot any possible mental health issues that can impair not only academic performance but also sleep quality and general life satisfaction.

- **Null Hypothesis (H_0):** Mean Stress Level ≤ 4 (The average stress level is at or below the "High Stress" benchmark)
- **Alternative Hypothesis (H_1):** Mean Stress Level > 4 (The average stress level exceeds the "High Stress" benchmark)
- **Significance Level (α):** 0.05
- **Test:** One-tailed one-sample t-test

```
> t_test_stress <- t.test(stress_level_clean, mu = 4, alternative = "greater", conf.level = 0.95)
> print(t_test_stress)

One Sample t-test

data: stress_level_clean
t = -2.6277, df = 29, p-value = 0.9932
alternative hypothesis: true mean is greater than 4
95 percent confidence interval:
 3.176688      Inf
sample estimates:
mean of x
      3.5
```

Figure 4.1.1.2: Hypothesis Testing: Stress Level

Conclusion

The second one-sample t-test yields statistically significant results. Because of the high p-value (0.9932), the null hypothesis is not rejected. This shows that there is no proof that the typical student population is under stress levels higher than the "High Stress" standard of 4.

With a sample mean of 3.5, which is exactly halfway on the 5-point scale, the cohort appears to be experiencing moderate levels of stress. The true average stress level for all students is moderate, according to the 95% confidence interval, which starts at 3.31. This result is noteworthy since it offers a measurable indicator of students' wellbeing. Although the average stress level is not extremely high, a mean of 3.5 indicates that stress is a common and regular occurrence for students. To keep it from getting worse, supporting

wellness programs and stress management tools should be used to address this mild but ongoing stress, which may have contributed to the previously noted performance gap in GPA.

4.1.2 Chi-Squared Testing

01. Activity Level and Sleep Quality are associated

To assess the relationship between students' self-reported sleep quality and their general level of activity, a simulation-based Chi-Squared test was employed. Since greater physical activity is often thought to have a good effect on sleep patterns, understanding this correlation is essential. This test investigates if there is a statistically significant correlation between the student body at SLIIT City Uni.

- **Null Hypothesis (H_0):** There is no association between activity level and sleep quality.
- **Alternative Hypothesis (H_1):** There is an association between activity level and sleep quality.
- **Significance Level (α):** 0.05
- **Test:** Chi-Squared test with Monte Carlo simulation (10,000 replicates)

Activity Level	Poor (1-2)	Average (3)	Good (4-5)	Total
Less Active	2	1	5	8
Active	5	6	5	16
Very Active	4	0	2	6
Total	11	7	12	30

Table 4.1.2.1: Contingency Table: Activity Level / Sleep Quality

```

> print(contingency_table)
      sleep_quality_cat
      Poor Average Good
Less active      2      1      5
Active           5      6      5
Very active      4      0      2
> result <- chisq.test(contingency_table, simulate.p.value = TRUE, B = 10000)
> print(result)

Pearson's Chi-squared test with simulated p-value (based on 10000 replicates)

data:  contingency_table
X-squared = 6.4617, df = NA, p-value = 0.1892

```

Figure 4.1.2.1: Chi Squared Testing: Activity Level / Sleep Quality

Conclusion

We are unable to reject the null hypothesis because the p-value (0.1892) is higher than the significance level of 0.05. At SLIIT City University, there is no statistically significant evidence to support a relationship between a student's perceived sleep quality and their overall activity level.

The data shows several intriguing tendencies, despite the fact that it lacks statistical significance. For example, nearly all of the "Very Active" students reported having trouble sleeping, yet none of them reported having average- quality sleep. On the other hand, the highest proportion of students who reported obtaining enough sleep belonged to the "Less Active" group. This surprising result raises the possibility that variables other than activity level, such as screen usage, mental exhaustion, or academic pressure, may have a greater impact on sleep quality in this student population. Thus, encouraging general physical activity on its own might not be a good way to help students sleep better; a more thorough plan that considers these additional elements is required.

02. Year of Study and Mental Exhaustion are associated

To ascertain whether there is a correlation between a student's year of study and the frequency of mental tiredness, a Chi-Squared test employing simulation was conducted. Academic pressure, workload, and future concerns frequently rise as students advance through their degrees. This test looks at whether

these difficulties show up as noticeably higher levels of mental tiredness in later life.

- **Null Hypothesis (H_0):** There is no association between year of study and mental exhaustion.
- **Alternative Hypothesis (H_1):** There is an association between year of study and mental exhaustion.
- **Significance Level (α):** 0.05
- **Test:** Chi-Squared test with Monte Carlo simulation (10,000 replicates)

Year of Study	Never	Not Often	Often	Total
Foundation	0	1	0	1
Year 1	1	2	1	4
Year 2	6	8	4	18
Year 3	3	3	1	7
Total	10	14	6	30

Table 4.1.2.2: Contingency Table: Year of Study/ Mental Exhaustion

```
> print(contingency_table)

      Never Not Often Often
Foundation    0      1     0
Year 1        1      2     1
Year 2        6      8     4
Year 3        3      3     1
> result <- chisq.test(contingency_table, simulate.p.value = TRUE, B = 10000)
> print(result)

Pearson's Chi-squared test with simulated p-value (based on 10000 replicates)

data: contingency_table
X-squared = 1.6757, df = NA, p-value = 1
```

Figure 4.1.2.2: Chi Squared Testing: Year of Study/ Mental Exhaustion

Conclusion

There is no statistically significant correlation between mental tiredness and the year of study, according to the findings of the Chi-Squared test ($X^2 = 1.6757$, $p = 1$). We are unable to rule out the null hypothesis because the p -value is higher than 0.05. This indicates that as students move from Foundation to Year 3, the prevalence of mental weariness does not routinely rise or fall. It's interesting to note that all year groups report feeling mentally fatigued, with a significant percentage of Year 2 kids stating they are "Often" exhausted. This implies that mental fatigue is a widespread problem that affects students at every stage of their academic careers, not only those in their last, usually more demanding years. Instead of being unique to a certain level of study, the underlying causes seem to be ingrained in the overall university experience itself, such as the never-ending cycle of assignments and tests. This emphasizes the necessity of providing all kids, regardless of year, with continuous, easily accessible mental health and wellbeing care.

Chapter 5 – Discussions and conclusions

5.1 Discussions of Key Findings

The study's conclusions offer important new information about how students' digital lifestyles, sleeping patterns, general wellbeing, and academic achievement are related. According to the demographic study, most of the respondents were in the School of Computing and were between the ages of 18 and 20. This implies that younger undergraduate students' experiences are largely represented in the study, which is consistent with the university's overall student body.

According to descriptive statistics, the majority of students report having modest amounts of sleep and poor quality, and they spend a significant portion of their days using screens for non-academic purposes. Additionally, the average stress level was neither low nor high, indicating the psychological challenges that university students frequently face. The fact that students report both good and bad sleep quality points to a variety of lifestyle patterns that may be related to individual variances in coping mechanisms, schedules, or academic loads. Some of these findings were confirmed by inferential analysis. While not all of the t-test results were statistically significant, they did reveal variations in GPA. Students with poor and good sleep quality showed significantly. The relationship between physical activity and stress level was further supported by chi-square tests, which also showed relationships between year of study and mental tiredness.

All things considered, these results are consistent with previous studies that show a connection between digital behavior, lifestyle choices, and academic performance. They also point to areas that require more research in a university setting.

5.2 Conclusion

The association between university students' screen time, sleep habits, academic achievement, and general well-being was investigated in this study. The results showed that, although there are discernible differences in their digital habits, the majority of students have moderate stress levels and medium sleep quality. Significant correlations between stress levels and sleep quality as well as between lifestyle characteristics and mental health were revealed by inferential analysis. According to the findings, students can improve their academic and psychological outcomes by getting enough sleep and using technology responsibly. Good sleeping practices, less screen time, and regular exercise may all contribute to students' overall university experience.

5.3 Recommendations

It is advised that kids maintain appropriate ratios of screen time, sleep, and exercise to support improved academic achievement and general health. Colleges and universities can play a big part in this by launching awareness campaigns about stress management, sleep hygiene, and digital wellbeing, for example, with the aid of counseling services. Stress and mental fatigue may be lessened by promoting regular exercise through easily accessible fitness centers and wellness activities. Additionally, academic staff might incorporate self-care and time management techniques into counseling or orientation sessions. Finally, to gain a deeper knowledge of how lifestyle factors affect student achievement, more research with bigger and more diverse samples is encouraged.

References

1. Internet usage and its effect on the lifestyle of university students (no date) ResearchGate. Available at: https://www.researchgate.net/publication/346716495_Internet_Usage_and_Its_Effect_on_the_Lifestyle_of_University_Students (Accessed: 12 October 2025).
2. Riboldi, I., et al. (2024) 'Understanding university students' perspectives towards digital tools for mental health support: A cross-country study', *Clinical Practice and Epidemiology in Mental Health*, 20(1), p. e17450179271467. Available at: <https://doi.org/10.2174/0117450179271467231231060255>
3. Banje, F.U. (no date) Azmiye YINAL. ERIC. Available at: <https://files.eric.ed.gov/fulltext/EJ1401792.pdf> (Accessed: 12 October 2025).
4. Wang, B. (2025) 'A comparative analysis of the impacts of digital lifestyles on the physical, mental, and social health among older internet users', *Humanities & Social Sciences Communications*, 12(1), pp. 1–8. Available at: <https://doi.org/10.1057/s41599-025-04490-y>
5. Tsukahara, S., et al. (2020) 'Association of eHealth literacy with lifestyle behaviors in university students: Questionnaire-based cross-sectional study', *Journal of Medical Internet Research*, 22(6), p. e18155. Available at: <https://doi.org/10.2196/18155>
6. Lusianawati, H., et al. (2023) 'The impact of digital lifestyles on young people's health: Social media abuse, physical inactivity and psychological impacts', *West Science Interdisciplinary Studies*, 1(09), pp. 773–782. Available at: <https://doi.org/10.58812/wsis.v1i09.211>
7. UOTP Marketing (2025) 'How does technology impact student learning outcomes?', University of the Potomac, 17 January. Available at: <https://potomac.edu/how-does-technology-impact-student-learning/> (Accessed: 12 October 2025).
8. Technology: The good, the bad, the ugly (2020) University of Arkansas Grantham. Available at: <https://www.uagrantham.edu/blog/technology-the-good-the-bad-the-ugly/> (Accessed: 12 October 2025).
9. The impact of social media on mental health in students (no date) Capital University. Available at: <https://www.capital.edu/admission-aid/college-readiness-resource-hub/mental-health-hub/the-impact-of-social-media-on-mental-health-in-students/> (Accessed: 12 October 2025).